

Hybrid Virtual-Physical Companion Prototype Ideas (Student Version)

1. Three Distinct Product Concepts

[DormPebble] — A palm-sized M5Stack-based device with a tiny screen (virtual pet display) that syncs with a mobile/desktop App via Bluetooth; AI adjusts the pet's mood based on user's phone/desktop activity (e.g., calm if user is studying, playful if scrolling social media) — [IoT+AI+App]

[DIY Buddy Hub] — An open-source kit (M5StickC Plus + basic circuit board) where students can 3D-print a custom case, program pet behaviors (via simple code), and sync pet progress across devices through a free Web tool — [IoT+AI+Web]

[GlowBuddy] — A USB-powered LED light (physical hardware) paired with a lightweight App/desktop widget; the virtual pet's state (hungry, happy, sleepy) controls the LED color, and pet activity syncs automatically between phone and laptop — [App+IoT]

2. Most Interesting Concept: DIY Buddy Hub

Who uses it: Chinese university students into maker/DIY culture, living in pet-restricted dorms, feeling lonely or wanting a creative outlet alongside emotional companionship.

When: Free weekends (to assemble/customize), study breaks (as a quiet companion), or group maker sessions (to show off custom pet designs).

What changes for them: They get a companion that feels unique (not pre-scripted), avoid the frustration of unsynced devices, and gain a low-stakes creative activity to reduce academic stress.

3. Physical Sensor/Hardware to Improve It

A low-cost (¥20-30) light sensor — it lets the pet adapt to the dorm's lighting (e.g., "sleeps" when the room is dark, becomes active when lights are on), making the companion feel more responsive and lifelike without complex setup.